

SOLIDWORKS API (.NET, VBA, C#) — Reference

(27/337): IBody2 (part 2 of 5)

SOLIDWORKS API Documentation (converted from CHM)

IBody2 (part 2 of 5) (Type)

SOLIDWORKS API Help

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DeleteFaces2 Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : DeleteFaces2 Method (IBody2)

Obsolete. Superseded by [IBody2::DeleteFaces3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function DeleteFaces2(  
    ByVal NumOfFaces As System.Integer, _  
    ByVal FaceList As System.Object, _  
    ByVal Option As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim NumOfFaces As System.Integer  
Dim FaceList As System.Object  
Dim Option As System.Integer  
Dim value As System.Integer  
  
value = instance.DeleteFaces2(NumOfFaces, FaceList, Option)
```

C#

```
System.int DeleteFaces2(  
    System.int NumOfFaces,  
    System.object FaceList,  
    System.int Option  
)
```

C++/CLI

```
System.int DeleteFaces2(  
& System.int NumOfFaces,  
& System.Object^ FaceList,  
& System.int Option  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NumOfFaces

FaceList

Option

Visual Basic for Applications (VBA) Syntax

See [Body2::DeleteFaces2](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

DeleteFaces3 Method (IBody2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : DeleteFaces3 Method (IBody2)

Obsolete. Superseded by [IBody2::DeleteFaces4](#).

.NET Syntax

Visual Basic (Declaration)

```
Function DeleteFaces3( _  
    ByVal FaceList As System.Object, _  
    ByVal Option As System.Integer, _  
    ByVal DoLocalCheck As System.Boolean, _  
    ByRef LocalCheckResult As System.Boolean _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim FaceList As System.Object  
Dim Option As System.Integer  
Dim DoLocalCheck As System.Boolean  
Dim LocalCheckResult As System.Boolean  
Dim value As System.Boolean  
  
value = instance.DeleteFaces3(FaceList, Option, DoLocalCheck, LocalCheckResult)
```

C#

```
System.bool DeleteFaces3(  
    System.object FaceList,  
    System.int Option,  
    System.bool DoLocalCheck,  
    out System.bool LocalCheckResult  
)
```

C++/CLI

```
System.bool DeleteFaces3(  
& System.Object^ FaceList,  
& System.int Option,  
& System.bool DoLocalCheck,  
& [Out] System.bool LocalCheckResult  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FaceList

Array containing the [faces](#) for deletion

Option

Additional control (see **Remarks**)

DoLocalCheck

True checks the bodies during the operation and sets the return value to indicate whether or not the resultant body is valid, false does not

LocalCheckResult

True if body is valid, false if not; to obtain this value, you must pass True for the DoLocalCheck argument

Return Value

True if a set of faces are deleted, false if not

Visual Basic for Applications (VBA) Syntax

See [Body2::DeleteFaces3](#).

Example

[Delete Faces \(VBA\)](#)

Remarks

All of the specified faces, which must belong to this temporary body, are deleted from IBody2. If the resulting body does not have a complete boundary, then SOLIDWORKS treats any holes as wounds and heals them as specified by the option argument. The option argument takes the following values:

Value	Loops on the faces to delete are...
0	Dependent on each other and should be healed at the same time. If extending faces does not yield a solution, then SOLIDWORKS tries to shrink the faces.
1	Independent and should be healed separately. This option also grows the parent faces around the hole to cover it.
2	Independent and should be healed separately. This option also finds a surface in which all edges of a hole lie and attaches this to a face covering the hole (SOLIDWORKS creates a new face to cover the hole).

For example, consider a cube with a through hole made up of four faces (a square hole). To delete these four faces, specify option 0 because the loop on the first face to be deleted is dependent on the loop of the second face to be deleted. Likewise, the loop on the second face to be deleted is dependent on the third face to be deleted, and so on.

Now consider the same cube with a through hole, except this through hole is a simple cylinder (one face). To delete the cylindrical face, specify option 1 to heal the loops independently. This is necessary because the cylindrical face actually has two loops (one at either end of the cylinder) that need to be healed separately.

It is possible to generate invalid geometry when you use this method because checking is disabled. Call [IBody2::Check3](#) to verify that the body is valid solid after using this method.

 See Also

[IBody2 Interface](#)
[IBody2 Members](#)

 Availability

SOLIDWORKS 2004 SP1, Revision Number 12

SOLIDWORKS API Help

[Send comments on this topic.](#)

DeleteFaces4 Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : DeleteFaces4 Method (IBody2)

Obsolete. Superseded by [IBody2::DeleteFaces5](#).

 .NET Syntax

Visual Basic (Declaration)

```
Function DeleteFaces4( _  
    ByVal FaceList As System.Object, _  
    ByVal Option As System.Integer, _  
    ByVal DoLocalCheck As System.Boolean, _  
    ByRef BodyList As System.Object, _  
    ByRef LocalCheckResult As System.Boolean _  
) As System.Boolean
```

Visual Basic (Usage)

```
Dim instance As IBody2  
Dim FaceList As System.Object  
Dim Option As System.Integer  
Dim DoLocalCheck As System.Boolean  
Dim BodyList As System.Object  
Dim LocalCheckResult As System.Boolean  
Dim value As System.Boolean  
  
value = instance.DeleteFaces4(FaceList, Option, DoLocalCheck, BodyList, LocalCheckResult)
```

C#

```
System.bool DeleteFaces4(  
    System.object FaceList,  
    System.int Option,  
    System.bool DoLocalCheck,  
    out System.object BodyList,
```

```
    out System.bool LocalCheckResult
)
```

C++/CLI

```
System.bool DeleteFaces4(
& System.Object^ FaceList,
& System.int Option,
& System.bool DoLocalCheck,
& [Out] System.Object^ BodyList,
& [Out] System.bool LocalCheckResult
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FaceList

Array containing the [faces](#) for deletion

Option

Additional control (see **Remarks**)

DoLocalCheck

True checks the bodies during the operation and sets the return value to indicate whether or not the resultant body is valid, false does not

BodyList

Array of temporary [bodies](#)

LocalCheckResult

True if body is valid, false if not; to obtain this value, you must pass True for the DoLocalCheck argument

Return Value

True if set of faces are deleted, false if not

Visual Basic for Applications (VBA) Syntax

See [Body2::DeleteFaces4](#).

Remarks

All of the specified faces, which must belong to this temporary body, are deleted from IBody2. If the resulting body does not have a complete boundary, then SOLIDWORKS treats any holes as wounds and heals them as specified by the option argument. The option argument takes the following values:

Value	Loops on the faces to delete are...
0	Dependent on each other and should be healed at the same time. If extending faces does not yield a solution, then SOLIDWORKS tries to shrink the faces.
1	Independent and should be healed separately. This option also grows the parent faces around the hole to cover it.
2	Independent and should be healed separately. This option also finds a surface in which all edges of a hole lie and attaches this to a face covering the hole (SOLIDWORKS creates a new face to cover the hole).

For example, consider a cube with a through hole made up of four faces (a square hole). To delete these four faces, specify option 0 because the loop on the first face to be deleted is dependent on the loop of the second face to be deleted. Likewise, the loop on the second face to be deleted is dependent on the third face to be deleted, and so on.

Now consider the same cube with a through hole, except this through hole is a simple cylinder (one face). To delete the cylindrical face, specify option 1 to heal the loops independently. This is necessary because the cylindrical face actually has two loops (one at either end of the cylinder) that need to be healed separately.

It is possible to generate invalid geometry when you use this method because checking is disabled. Call [IBody2::Check3](#) to verify that the body is valid solid after using this method.

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::DeleteBlends2 Method](#)
[IBody2::!DeleteBlends2 Method](#)
[IBody2::EnumFaces Method](#)
[IBody2::GetFirstFace Method](#)
[IBody2::!GetFirstFace Method](#)
[IBody2::!GetFaces Method](#)
[IBody2::GetFaces Method](#)
[IBody2::GetFaceCount Method](#)

Availability

DeleteFaces5 Method (IBody2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : DeleteFaces5 Method (IBody2)

Deletes a set of faces from a temporary body.

.NET Syntax

Visual Basic (Declaration)

```

Function DeleteFaces5( _
    ByVal FaceList As System.Object, _
    ByVal HealAction As System.Integer, _
    ByVal LoopProcessOption As System.Integer, _
    ByVal DoLocalCheck As System.Boolean, _
    ByRef BodyList As System.Object, _
    ByRef LocalCheckResult As System.Boolean _
) As System.Boolean

```

Visual Basic (Usage)



```

Dim instance As IBody2
Dim FaceList As System.Object
Dim HealAction As System.Integer
Dim LoopProcessOption As System.Integer
Dim DoLocalCheck As System.Boolean
Dim BodyList As System.Object
Dim LocalCheckResult As System.Boolean
Dim value As System.Boolean

value = instance.DeleteFaces5(FaceList, HealAction, LoopProcessOption, DoLocalCheck, BodyList, LocalCheckResult)

```

C#

```

System.bool DeleteFaces5(
    System.object FaceList,
    System.int HealAction,
    System.int LoopProcessOption,
    System.bool DoLocalCheck,
    out System.object BodyList,
    out System.bool LocalCheckResult
)

```

C++/CLI

```

System.bool DeleteFaces5(
    & System.Object^ FaceList,
    & System.int HealAction,
    & System.int LoopProcessOption,
    & System.bool DoLocalCheck,
    [Out] System.Object^ BodyList,
    [Out] System.bool LocalCheckResult
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FaceList

Array containing the [faces](#) for deletion

HealAction

Healing action as defined in [swHealActionType_e](#) (see **Remarks**)

LoopProcessOption

Loop processing as defined in [swLoopProcessOption_e](#) (see **Remarks**)

DoLocalCheck

True checks the bodies during the operation and sets the return value to indicate whether the resultant body is valid, false does not

BodyList

Array of temporary [bodies](#)

LocalCheckResult

True if body is valid, false if not; to obtain this value, you must pass true for the DoLocalCheck argument

Return Value

True if set of faces are deleted, false if not

Visual Basic for Applications (VBA) Syntax

See [Body2::DeleteFaces5](#).

Example

[Create New Part from Existing Part Using Temporary Body \(VB.NET\)](#)

[Create New Part from Existing Part Using Temporary Body \(VBA\)](#)

[Create New Part from Existing Part Using Temporary Body \(C#\)](#)

Remarks

When faces are deleted, wounds are created on those faces that require healing. Healing can be accomplished by extending neighboring faces, shrinking the deleted faces, or covering the wounds. The value that you specify for HealAction determines how wounds are healed.

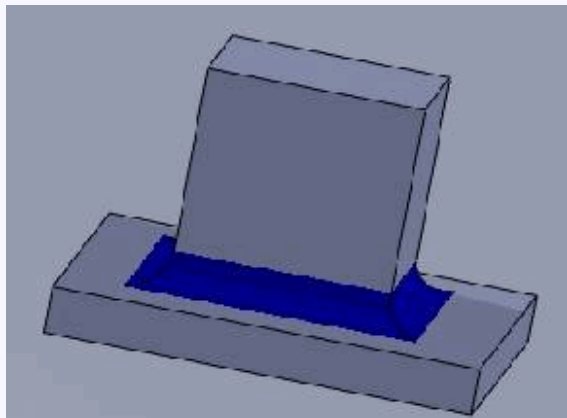
When a wound has multiple loops, you can tell SOLIDWORKS how to process the loops individually or together, or you can let SOLIDWORKS decide how to process the loops. The value you specify for LoopProcessAction determines how to process multiple loops.

For example, consider a cube with a through hole made up of four faces (a square hole). To delete these four faces, specify `swHealAction_Shrink` for HealAction and `swLoopProcess_Together` for LoopProcessAction because the loop on the first face to be deleted is dependent on the loop of the second face to be deleted. Likewise, the loop on the second face to be deleted is dependent on the third face to be deleted, and so on.

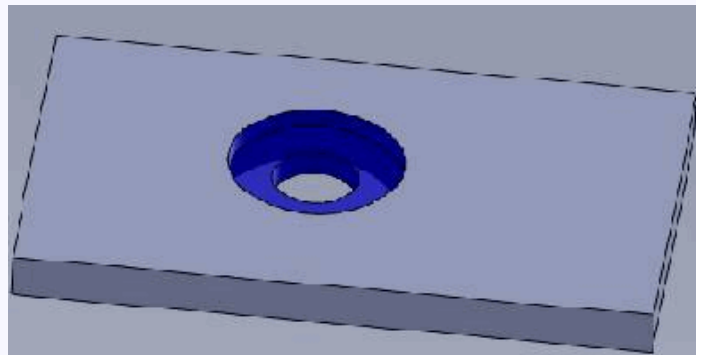
Now consider the same cube with a through hole, except this through hole is a simple cylinder (one face). To delete the cylindrical face, specify `swHealActionShrink_e` for HealAction and `swLoopProcess_Independent` for LoopProcessAction to heal the loops independently. This is necessary because the cylindrical face actually has two loops (one at either end of the cylinder) that need to be healed separately.

It is possible to generate invalid geometry when you use this method because checking is disabled. Call [IBody2::Check3](#) to verify that the body is valid solid after using this method.

swLoopProcess_Together



swLoopProcess_Independent



See Also

[IBody2 Interface](#)

[IBody2 Members](#)

[IBody2::DeleteBlends3 Method](#)

[IBody2::DeleteBlends3 Method](#)

[IBody2::EnumFaces Method](#)

[IBody2::GetFaceCount Method](#)

[IBody2::GetFirstFace Method](#)

[IBody2::IGetFaces Method](#)

[IBody2::IGetFirstFace Method](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

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DeleteFacesMakeSheetBodies Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : DeleteFacesMakeSheetBodies Method (IBody2)

Creates sheet bodies from deleted faces.

.NET Syntax

Visual Basic (Declaration)

```
Function DeleteFacesMakeSheetBodies( _  
    ByVal FaceList As System.Object _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim FaceList As System.Object  
Dim value As System.Object  
  
value = instance.DeleteFacesMakeSheetBodies(FaceList)
```

C#

```
System.object DeleteFacesMakeSheetBodies(  
    System.object FaceList  
)
```

C++/CLI

```
System.Object^ DeleteFacesMakeSheetBodies(  
& System.Object^ FaceList  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FaceList
Array of [faces](#) to delete

Return Value

Array of sheet [bodies](#) created from the deleted faces

Visual Basic for Applications (VBA) Syntax

See [Body2::DeleteFacesMakeSheetBodies](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::DeleteFacesMakeSheetBodies Method](#)
[IFace2::CreateSheetBody Method](#)
[IFace2::CreateSheetBodyByFaceExtension Method](#)
[IModeler::CreateSheetFromFaces Method](#)
[IModeler::CreateSheetFromSurface Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12

SOLIDWORKS API Help

[Send comments](#) on this topic.

DeSelect Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : DeSelect Method (IBody2)

Deselects this body.

.NET Syntax

Visual Basic (Declaration)

```
Function DeSelect() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim value As System.Boolean  
  
value = instance.DeSelect()
```

C#

```
System.bool DeSelect()
```

C++/CLI

```
System.bool DeSelect();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if the body was deselected, false if it was not

Visual Basic for Applications (VBA) Syntax

See [Body2::DeSelect](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 10

SOLIDWORKS API Help

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Diagnose Method (IBody2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : Diagnose Method (IBody2)

Gets the [IDiagnoseResult](#) object for this body.

.NET Syntax

Visual Basic (Declaration)

```
Function Diagnose() As DiagnoseResult
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim value As DiagnoseResult  
  
value = instance.Diagnose()
```

C#

```
DiagnoseResult Diagnose()
```

C++/CLI

```
DiagnoseResult^ Diagnose();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Pointer to the [IDiagnoseResult](#) object

Visual Basic for Applications (VBA) Syntax

See [Body2::Diagnose](#).

Example

[Get and Fill Gaps in Body \(C#\)](#)
[Get and Fill Gaps in Body \(VB.NET\)](#)
[Get and Fill Gaps in Body \(VBA\)](#)

Remarks

Use the [IDiagnoseResult](#) object to get the gaps and coedges in each gap on this body.

See Also

[IBody2 Interface](#)
[IBody2 Members](#)

Availability

SOLIDWORKS 2004 SP4, Revision Number 12

SOLIDWORKS API Help

[Send comments on this topic.](#)

DisableDisplay Property (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : DisableDisplay Property (IBody2)

Gets or sets whether to hide or show this body.

.NET Syntax

Visual Basic (Declaration)

Property DisableDisplay As System.Boolean

Visual Basic (Usage)



```
Dim instance As IBody2
Dim value As System.Boolean

instance.DisableDisplay = value

value = instance.DisableDisplay
```

C#

```
System.bool DisableDisplay {get; set;}
```

C++/CLI

```
property System.bool DisableDisplay {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to hide the body; false to show the body

NOTE: If true, then the body is hidden but remains selectable.

Visual Basic for Applications (VBA) Syntax

See [Body2::DisableDisplay](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::HideBody Method](#)
[IModelDoc2::HideShowBodies Method](#)

DisableHighlight Property (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : DisableHighlight Property (IBody2)

Disables highlighting of the selected body in the graphics area.

.NET Syntax

Visual Basic (Declaration)

```
Property DisableHighlight As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim value As System.Boolean  
  
instance.DisableHighlight = value  
value = instance.DisableHighlight
```

C#

```
System.bool DisableHighlight {get; set;}
```

C++/CLI

```
property System.bool DisableHighlight {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to disable highlighting, false to not

Visual Basic for Applications (VBA) Syntax

Visual Basic for Applications (VBA)

```
'-----  
'  
' Preconditions: Model document is open and  
'               contains at least one body.  
'  
' Postconditions: None  
'  
'-----
```

Option Explicit

```
Dim swApp As SldWorks.SldWorks  
Dim swPart As SldWorks.PartDoc  
Dim swBody As SldWorks.Body2  
Dim swModel As SldWorks.ModelDoc2  
Dim swModelDocExt As SldWorks.ModelDocExtension  
Dim vBodies As Variant  
Dim i As Long  
Dim sBodySelStr As String, sBodyTypeSelStr As String  
Dim bRet As Boolean
```

```

Sub main()

Set swApp = Application.SldWorks
Set swPart = swApp.ActiveDoc
Set swModel = swPart

vBodies = swPart.GetBodies2(swAllBodies, True)
If IsEmpty(vBodies) Then End

Set swModelDocExt = swModel.Extension

For i = 0 To UBound(vBodies)
    Set swBody = vBodies(i)
    sBodySelStr = swBody.GetSelectionId
    Select Case swBody.GetType
        Case swSolidBody
            sBodyTypeSelStr = "SOLIDBODY"
        Case swSheetBody
            sBodyTypeSelStr = "SURFACEBODY"
        Case Else
            Debug.Assert False
    End Select

    ' Select a body
    bRet = swModelDocExt.SelectByID2(sBodySelStr, sBodyTypeSelStr, 0#, 0#, 0#, True, 0, Nothing, swSelectOptionDefault): Debug.Assert bRet
    ' Disable highlighting of the body
    swBody.DisableHighlight = True

    ' Select the body again to check that highlighting is off
    swModel.ClearSelection2 True
    bRet = swModelDocExt.SelectByID2(sBodySelStr, sBodyTypeSelStr, 0#, 0#, 0#, True, 0, Nothing, swSelectOptionDefault): Debug.Assert bRet
    ' Re-enable highlighting of the body
    swBody.DisableHighlight = False

    ' Select the body again to check that highlighting is on
    swModel.ClearSelection2 True
    bRet = swModelDocExt.SelectByID2(sBodySelStr, sBodyTypeSelStr, 0#, 0#, 0#, True, 0, Nothing, swSelectOptionDefault): Debug.Assert bRet

    swModel.ClearSelection2 True

Next i

End Sub

```

See Also

[IBody2 Interface](#)
[IBody2 Members](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

Display Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : Display Method (IBody2)

Obsolete. Superseded by [IBody2::Display3](#).

.NET Syntax

Visual Basic (Declaration)

```

Sub Display( _
    ByVal Part As System.Object, _
    ByVal Color As System.Integer _
)

```

```
Dim instance As IBody2
Dim Part As System.Object
Dim Color As System.Integer

instance.Display(Part, Color)
```

```
void Display(
    System.object Part,
    System.int Color
)
```

```
void Display(
    & System.Object^ Part,
    & System.int Color
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Part

Color

Visual Basic for Applications (VBA) Syntax

See [Body2::Display](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)

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Display2 Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : Display2 Method (IBody2)

Obsolete. Superseded by [IBody2::Display3](#).

.NET Syntax

```
Sub Display2( _
    ByVal Part As PartDoc, _
    ByVal Color As System.Integer, _
    ByVal Option As System.Integer _
)
```

```
Dim instance As IBody2
Dim Part As PartDoc
Dim Color As System.Integer
Dim Option As System.Integer

instance.Display2(Part, Color, Option)
```

```
void Display2(
    PartDoc Part,
    System.int Color,
```

```
System.int Option
)
```

C++/CLI

```
void Display2(
& PartDoc^ Part,
& System.int Color,
& System.int Option
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Part

Color

Option

Visual Basic for Applications (VBA) Syntax

See [Body2::Display2](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

Display3 Method (IBody2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : Display3 Method (IBody2)

Displays this temporary body in the context of the specified part or component.

NET Syntax

Visual Basic (Declaration)

```
Function Display3( _
    ByVal Component As System.Object, _
    ByVal Color As System.Integer, _
    ByVal Option As System.Integer _
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IBody2
Dim Component As System.Object
Dim Color As System.Integer
Dim Option As System.Integer
Dim value As System.Integer

value = instance.Display3(Component, Color, Option)
```

C#

```
System.int Display3(
    System.object Component,
    System.int Color,
    System.int Option
)
```

C++/CLI

```
System.int Display3(
& System.Object^ Component,
& System.int Color,
```

```
& System.int Option  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Component

Part or component where the temporary body exists (see **Remarks**)

Color

COLORREF value (see **Remarks**)

Option

Selection state of temporary body as defined by [swTempBodySelectOptions_e](#) (see **Remarks**)

Return Value

- 0 = Success
- 1 = Failed because this body is not a temporary body
- 2 = Invalid component
- 3 = Not a part instance

Visual Basic for Applications (VBA) Syntax

See [Body2::Display3](#).

Example

[Create Loft Body \(VB.NET\)](#)
[Create Loft Body \(VBA\)](#)
[Create Loft Body \(C#\)](#)
[Display Temporary Body \(C#\)](#)
[Display Temporary Body \(VB.NET\)](#)
[Display Temporary Body \(VBA\)](#)

Remarks

This method:

- Is valid only for temporary bodies. To determine whether a body is temporary, use [IBody2::IsTemporaryBody](#).
- Applies Color to this temporary body in the graphics area, effectively selecting it.

You can also use [IBody2::MaterialPropertyValues2](#) to change the appearance of this temporary body.

Component cannot be in a subassembly.

Specifying Option with swTempBodySelectable sets the blocking state to [swBlockingStates_e.swModifyBlock](#). Unset the blocking state by calling [IBody2::Hide](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15

SOLIDWORKS API Help

[Send comments](#) on this topic.

DisplayWireFrameXOR Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : DisplayWireFrameXOR Method (IBody2)

Displays a temporary body in the given part's context in XOR mode.

.NET Syntax

Visual Basic (Declaration)

```
Sub DisplayWireFrameXOR(  
    ByVal Part As System.Object, _  
    ByVal Color As System.Integer _  
)
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim Part As System.Object  
Dim Color As System.Integer  
  
instance.DisplayWireFrameXOR(Part, Color)
```

C#

```
void DisplayWireFrameXOR(  
    System.object Part,  
    System.int Color  
)
```

C++/CLI

```
void DisplayWireFrameXOR(  
& System.Object^ Part,  
& System.int Color  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Part

Part

Color

Color of part

Visual Basic for Applications (VBA) Syntax

See [Body2::DisplayWireFrameXOR](#).

See Also

[IBody2 Interface](#)

[IBody2 Members](#)

[IBody2::DisplayWireFrameXOR Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

DraftBody Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : DraftBody Method (IBody2)

Obsolete. Superseded by [IBody2::DraftBody2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function DraftBody( _  
    ByVal NumOfFaces As System.Integer, _  
    ByVal FaceList As System.Object, _  
    ByVal EdgeList As System.Object, _  
    ByVal DraftAngle As System.Double, _  
    ByVal Dir As System.Object _  
) As System.Boolean
```

Visual Basic (Usage)

```
Dim instance As IBody2  
Dim NumOfFaces As System.Integer  
Dim FaceList As System.Object  
Dim EdgeList As System.Object  
Dim DraftAngle As System.Double
```



```
Dim Dir As System.Object
Dim value As System.Boolean

value = instance.DraftBody(NumOfFaces, FaceList, EdgeList, DraftAngle, Dir)
```

C#

```
System.bool DraftBody(
    System.int NumOfFaces,
    System.object FaceList,
    System.object EdgeList,
    System.double DraftAngle,
    System.object Dir
)
```

C++/CLI

```
System.bool DraftBody(
    & System.int NumOfFaces,
    & System.Object^ FaceList,
    & System.Object^ EdgeList,
    & System.double DraftAngle,
    & System.Object^ Dir
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NumOfFaces

FaceList

EdgeList

DraftAngle

Dir

Visual Basic for Applications (VBA) Syntax

See [Body2::DraftBody](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

DraftBody2 Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : DraftBody2 Method (IBody2)

Adds drafts to the specified faces on a temporary body. This method modifies the body.

.NET Syntax

Visual Basic (Declaration)

```
Function DraftBody2( _
    ByVal NumOfFaces As System.Integer, _
    ByVal FaceList As System.Object, _
    ByVal EdgeList As System.Object, _
    ByVal BasePoint As System.Object, _
    ByVal DraftAngle As System.Double, _
    ByVal Dir As System.Object _
) As System.Boolean
```

Visual Basic (Usage)

```
Dim instance As IBody2
Dim NumOfFaces As System.Integer
```



```

Dim FaceList As System.Object
Dim EdgeList As System.Object
Dim BasePoint As System.Object
Dim DraftAngle As System.Double
Dim Dir As System.Object
Dim value As System.Boolean

value = instance.DraftBody2(NumOfFaces, FaceList, EdgeList, BasePoint, DraftAngle, Dir)

```

C#

```

System.bool DraftBody2(
    System.int NumOfFaces,
    System.object FaceList,
    System.object EdgeList,
    System.object BasePoint,
    System.double DraftAngle,
    System.object Dir
)

```

C++/CLI

```

System.bool DraftBody2(
    & System.int NumOfFaces,
    & System.Object^ FaceList,
    & System.Object^ EdgeList,
    & System.Object^ BasePoint,
    & System.double DraftAngle,
    & System.Object^ Dir
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NumOfFaces

Number of faces to draft

FaceList

Array of [faces](#) to draft

EdgeList

Array of [edges](#), one for each face, along which to apply the drafts

BasePoint

Pointer to a [MathPoint](#) object (x,y,z values of the base point)

DraftAngle

Draft angle

Dir

Array of 3 doubles (x, y, z), a vector which specifies the direction of the draft

Return Value

True if drafts are applied, false if not

Visual Basic for Applications (VBA) Syntax

See [Body2::DraftBody2](#).

See Also

[IBody2 Interface](#)

[IBody2 Members](#)

[IBody2::IDraftBody2 Method](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14

SOLIDWORKS API Help

[Send comments on this topic.](#)

EnumFaces Method (IBody2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : EnumFaces Method (IBody2)

Returns an enumerated list of the faces in a body.

.NET Syntax

Visual Basic (Declaration)

```
Function EnumFaces() As EnumFaces2
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim value As EnumFaces2  
  
value = instance.EnumFaces()
```

C#

```
EnumFaces2 EnumFaces()
```

C++/CLI

```
EnumFaces2^ EnumFaces();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Enumerated list of [faces](#) (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Body2::EnumFaces](#).

Example

[Enumerate Bodies \(C++\)](#)

Remarks

OLE automation applications should use [IBody2::GetFirstFace](#) and [IFace2::GetNextFace](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ExtendSurface Method (IBody2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : ExtendSurface Method (IBody2)

Creates a new temporary body by extending the selected edges.

.NET Syntax

Visual Basic (Declaration)

```
Function ExtendSurface( _  
    ByVal EdgesToExtend As System.Object, _  
    ByVal ExtendLinear As System.Boolean, _  
    ByVal EndCondition As System.Integer, _  
    ByVal Dist As System.Double, _  
    ByVal PUpToVtx As Vertex, _  
    ByVal PUpToFace As Face _  
) As Body2
```

```

Dim instance As IBody2
Dim EdgesToExtend As System.Object
Dim ExtendLinear As System.Boolean
Dim EndCondition As System.Integer
Dim Dist As System.Double
Dim PUpToVtx As Vertex
Dim PUpToFace As Face
Dim value As Body2

```

```
value = instance.ExtendSurface(EdgesToExtend, ExtendLinear, EndCondition, Dist, PUpToVtx, PUpToFace)
```

C#

```

Body2 ExtendSurface(
    System.object EdgesToExtend,
    System.bool ExtendLinear,
    System.int EndCondition,
    System.double Dist,
    Vertex PUpToVtx,
    Face PUpToFace
)

```

C++/CLI

```

Body2^ ExtendSurface(
    & System.Object^ EdgesToExtend,
    & System.bool ExtendLinear,
    & System.int EndCondition,
    & System.double Dist,
    & Vertex^ PUpToVtx,
    & Face^ PUpToFace
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

EdgesToExtend

Array of the selected edges to extend

ExtendLinear

True to extend the selected edges linearly along the tangent of the face at the edges; false to extend the selected edges in the same direction as the face

EndCondition

- 0 = Extend selected edges per value specified for Dist

- 1 = Extend selected edges to PUpToVtx
- 2 = Extend selected edges to PUpToFace

Dist

Distance by which to extend the selected edges (see **Remarks**)

PUpToVtx

[Vertex](#) up to which to extend the selected edges (see **Remarks**)

PUpToFace

[Face](#) up to which to extend the selected edges (see **Remarks**)

Return Value

Pointer to the newly created [IBody2](#) object

Visual Basic for Applications (VBA) Syntax

See [Body2::ExtendSurface](#).

Example

[Extend Surface \(VBA\)](#)

Remarks

This method supports surface bodies only.

You can extend the edges by a specified distance or up to a vertex or up to a face.

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::IExtendSurface Method](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14

SOLIDWORKS API Help

[Send comments on this topic.](#)

FindAttribute Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : FindAttribute Method (IBody2)

Finds an attribute on a body.

.NET Syntax

Visual Basic (Declaration)

```
Function FindAttribute( _  
    ByVal AttributeDef As AttributeDef, _  
    ByVal WhichOne As System.Integer _  
) As Attribute
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim AttributeDef As AttributeDef  
Dim WhichOne As System.Integer  
Dim value As Attribute  
  
value = instance.FindAttribute(AttributeDef, WhichOne)
```

C#

```
Attribute FindAttribute(  
    AttributeDef AttributeDef,  
    System.int WhichOne  
)
```

C++/CLI

```
Attribute^ FindAttribute(  
& AttributeDef^ AttributeDef,  
& System.int WhichOne  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

AttributeDef

Pointer to the [attribute definition](#) that you want to find

WhichOne

Instance number of this type on the body (0, 1, 2, 3, and so on)

Return Value

Pointer to the [attribute](#) instance

Visual Basic for Applications (VBA) Syntax

See [Body2::FindAttribute](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetBoundingBox Method (IBody2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : GetBoundingBox Method (IBody2)

Gets the bounding box for this body.

.NET Syntax

Visual Basic (Declaration)

```
Function GetBoundingBox() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IBody2
Dim value As System.Object

value = instance.GetBoundingBox()
```

C#

```
System.object GetBoundingBox()
```

C++/CLI

```
System.Object^ GetBoundingBox();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of 6 doubles representing the extents of the bounding box (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Body2::GetBoundingBox](#).

Example

[Cut Body in Half Using Macro Feature \(VBA\)](#)

Remarks

The X,Y,Z points returned by SOLIDWORKS are the lower and upper diagonal corners which bound the body with the box sides parallel to the X, and Z axes. The box dimensions returned by SOLIDWORKS enclose the body and are typically close to the minimum possible size (this is typical but not always true).

The return value is an array of doubles as follows:

[XCorner1, YCorner1, ZCorner1, XCorner2, YCorner2, ZCorner2]

IMPORTANT: The values returned are approximate and should not be used for comparison or calculation purposes. Furthermore, the bounding box may vary after rebuilding the model.

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::IGetBoundingBox Method](#)
[IAssemblyDoc::GetBox Method](#)
[IAssemblyDoc::IGetBox Method](#)
[IComponent2::GetBox Method](#)
[IComponent2::IGetBox Method](#)
[IFace2::GetBox Method](#)
[IFace2::IGetBox Method](#)
[IFeature::GetBox Method](#)
[IFeature::IGetBox Method](#)
[IModelDocExtension::GetVisibleBox Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetCoincidenceTransform Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : GetCoincidenceTransform Method (IBody2)

Obsolete. Superseded by [IBody2::GetCoincidenceTransform2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetCoincidenceTransform( _  
    ByVal BodyDispIn As System.Object, _  
    ByRef Xform As MathTransform _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim BodyDispIn As System.Object  
Dim Xform As MathTransform  
Dim value As System.Boolean  
  
value = instance.GetCoincidenceTransform(BodyDispIn, Xform)
```

C#

```
System.bool GetCoincidenceTransform(  
    System.object BodyDispIn,  
    out MathTransform Xform  
)
```

C++/CLI

```
System.bool GetCoincidenceTransform(  
& System.Object^ BodyDispIn,  
& [Out] MathTransform^ Xform  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

BodyDispIn
Input body

Xform
Pointer to the [transformation matrix](#)

Return Value

True if the bodies (the body on which the method is invoked and the input body) can coincide by applying appropriate transforms, false if not

Visual Basic for Applications (VBA) Syntax

See [Body2::GetCoincidenceTransform](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)

Availability

GetCoincidenceTransform2 Method (IBody2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : GetCoincidenceTransform2 Method (IBody2)

Calculates the transformation matrix, which would make the input body coincident with this body if the transformation matrix is applied.

.NET Syntax

Visual Basic (Declaration)

```
Function GetCoincidenceTransform2( _
    ByVal BodyDispIn As System.Object, _
    ByRef Xform As MathTransform _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IBody2
Dim BodyDispIn As System.Object
Dim Xform As MathTransform
Dim value As System.Boolean

value = instance.GetCoincidenceTransform2(BodyDispIn, Xform)
```

C#

```
System.bool GetCoincidenceTransform2(
    System.object BodyDispIn,
    out MathTransform Xform
)
```

C++/CLI

```
System.bool GetCoincidenceTransform2(
    & System.Object^ BodyDispIn,
    [Out] MathTransform^ Xform
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

BodyDispIn

[Input body](#).

Xform

Pointer to the [transformation matrix](#)

Return Value

True if this body and the input body can coincide by applying the transformation matrix, false if not

Visual Basic for Applications (VBA) Syntax

See [Body2::GetCoincidenceTransform2](#).

Example

[Calculate Transformations in Part \(C#\)](#)

[Calculate Transformations in Part \(VB.NET\)](#)

[Calculate Transformations in Part \(VBA\)](#)

See Also

[IBody2 Interface](#)

[IBody2 Members](#)

Availability

GetEdgeCount Method (IBody2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : GetEdgeCount Method (IBody2)

Gets the number of edges for this body.

.NET Syntax

Visual Basic (Declaration)

```
Function GetEdgeCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IBody2
Dim value As System.Integer

value = instance.GetEdgeCount()
```

C#

```
System.int GetEdgeCount()
```

C++/CLI

```
System.int GetEdgeCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of edges

Visual Basic for Applications (VBA) Syntax

See [Body2::GetEdgeCount](#).

Example

[Get Original Body from Pattern Body \(VBA\)](#)
[Get Original Body from Pattern Body \(VB.NET\)](#)
[Get Original Body from Pattern Body \(C#\)](#)

Remarks

Call this method before calling [IBody2::IGetEdges](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

GetEdges Method (IBody2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : GetEdges Method (IBody2)

Gets the edges for this body.

.NET Syntax

Visual Basic (Declaration)

```
Function GetEdges() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IBody2
Dim value As System.Object

value = instance.GetEdges()
```

C#

```
System.object GetEdges()
```

C++/CLI

```
System.Object^ GetEdges();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of [edges](#)

Visual Basic for Applications (VBA) Syntax

See [Body2::GetEdges](#).

Example

[Cut Body in Half Using Macro Feature \(VBA\)](#)
[Display Vertices \(VBA\)](#)
[Check Edges for Faults \(C#\)](#)
[Check Edges for Faults \(VB.NET\)](#)
[Check Edges for Faults \(VBA\)](#)

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::GetEdges Method](#)
[IModelDocExtension::SelectAll Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetExcessBodyArray Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : GetExcessBodyArray Method (IBody2)

Gets the excess bodies after sewing.

.NET Syntax

Visual Basic (Declaration)

```
Function GetExcessBodyArray() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IBody2
Dim value As System.Object

value = instance.GetExcessBodyArray()
```

C#

```
System.object GetExcessBodyArray()
```

C++/CLI

```
System.Object^ GetExcessBodyArray();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of excess [bodies](#)

Visual Basic for Applications (VBA) Syntax

See [Body2::GetExcessBodyArray](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::IGetExcessBodyArray Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetExtremePoint Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : GetExtremePoint Method (IBody2)

Calculates the extreme point of the model in the specified direction.

.NET Syntax

Visual Basic (Declaration)

```
Function GetExtremePoint( _  
    ByVal X As System.Double, _  
    ByVal Y As System.Double, _  
    ByVal Z As System.Double, _  
    ByRef Outx As System.Double, _  
    ByRef Outy As System.Double, _  
    ByRef Outz As System.Double _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim X As System.Double  
Dim Y As System.Double  
Dim Z As System.Double  
Dim Outx As System.Double  
Dim Outy As System.Double  
Dim Outz As System.Double  
Dim value As System.Boolean  
  
value = instance.GetExtremePoint(X, Y, Z, Outx, Outy, Outz)
```

C#

```
System.bool GetExtremePoint(  
    System.double X,  
    System.double Y,  
    System.double Z,  
    out System.double Outx,  
    out System.double Outy,  
    out System.double Outz  
)
```

C++/CLI

```

System.bool GetExtremePoint(
& System.double X,
& System.double Y,
& System.double Z,
& [Out] System.double Outx,
& [Out] System.double Outy,
& [Out] System.double Outz
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

X
 X component of the direction vector
Y
 Y component of the direction vector
Z
 Z component of the direction vector
Outx
 Extreme point X coordinate
Outy
 Extreme point Y coordinate
Outz
 Extreme point Z coordinate

Return Value

True if a point was found, false if not

Visual Basic for Applications (VBA) Syntax

See [Body2::GetExtremePoint](#).

Remarks

This method returns the furthest possible point of intersection between a plane normal to the direction vector specified and the model as the plane moves along the direction vector. For example, if the model is a right cube centered on the origin and the direction vector is (1.0, 1.0, 1.0), then the extreme point is the vertex at (1.0, 1.0, 1.0).

If there is more than one point (for example, if there is a face perpendicular to the direction vector), SOLIDWORKS returns a unique point that it finds in a deterministic way.

COM example

```

HRESULT auBody_c::XDispatch2::GetExtremePoint( double x, double y, double z, double*
outx, double* outy, double* outz, VARIANT_BOOL* found ) {
    METHOD_PROLOGUE_EX_(auBody_c, Dispatch2)
    AU_INTERFACE_VERIFY_NOT_DISCONNECTED
    BOOL gotIt = pThis->GetExtremePoint(x, y, z, outx, outy, outz);
    *found = gotIt ? VARIANT_True : VARIANT_false;
    return gotIt ? S_OK : S_false;
}

```

See Also

[IBody2 Interface](#)
[IBody2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetFaceCount Method (IBody2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : GetFaceCount Method (IBody2)

Gets the number of faces in this body.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function GetFaceCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim value As System.Integer  
  
value = instance.GetFaceCount()
```

C#

```
System.int GetFaceCount()
```

C++/CLI

```
System.int GetFaceCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of faces

Visual Basic for Applications (VBA) Syntax

See [Body2::GetFaceCount](#).

Example

[Process Body \(C#\)](#)
[Process Body \(VB.NET\)](#)
[Process Body \(VBA\)](#)
[Fill Holes in Part \(C#\)](#)
[Fill Holes in Part \(VB.NET\)](#)
[Fill Holes in Part \(VBA\)](#)

Remarks

Call this method before calling [IBody2::IGetFaces](#) to determine the size of the array for that method.

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::GetFirstFace Method](#)
[IBody2::IGetFirstFace Method](#)
[IBody2::EnumFaces Method](#)
[IBody2::DeleteFaces3 Method](#)
[IBody2::IDeleteBlends2 Method](#)
[IBody2::DeleteBlends2 Method](#)
[IBody2::GetFaces Method](#)
[IBody2::IGetFaces Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetFaces Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : GetFaces Method (IBody2)

Gets all of the faces on the body.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function GetFaces() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim value As System.Object  
  
value = instance.GetFaces()
```

C#

```
System.object GetFaces()
```

C++/CLI

```
System.Object^ GetFaces();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of [faces](#) on the body

Visual Basic for Applications (VBA) Syntax

See [Body2::GetFaces](#).

Remarks

You can use this method instead of using [IBody2::GetFirstFace](#) and [IFace2::GetNextFace](#) to get all of the faces on a body.

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::IGetFaces Method](#)
[IBody2::GetFaceCount Method](#)
[IBody2::IGetFirstFace Method](#)
[IBody2::GetFirstFace Method](#)

Availability

SOLIDWORKS 2008 SP1, Revision Number 16.1

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetFeatureCount Method (IBody2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : GetFeatureCount Method (IBody2)

Gets the number of features in this body in a multibody sheet metal part.

NET Syntax

Visual Basic (Declaration)

```
Function GetFeatureCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim value As System.Integer  
  
value = instance.GetFeatureCount()
```

C#

```
System.int GetFeatureCount()
```

C++/CLI

```
System.int GetFeatureCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of features in this body

Visual Basic for Applications (VBA) Syntax

See [Body2::GetFeatureCount](#).

Example

[Get Features of Multibody Sheet Metal Part \(C#\)](#)
[Get Features of Multibody Sheet Metal Part \(VB.NET\)](#)
[Get Features of Multibody Sheet Metal Part \(VBA\)](#)

Remarks

The features of a body in a multibody sheet metal part are located in the solid bodies folder in the FeatureManager design tree.

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::GetFeatures Method](#)
[IBody2::IGetFeatures Method](#)
[IBodyFolder Interface](#)
[IBodyFolder::GetBodies Method](#)
[IBodyFolder::GetBodyCount Method](#)
[IBodyFolder::IGetBodies Method](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetFeatures Method (IBody2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : GetFeatures Method (IBody2)

Gets the features in this body in a multibody sheet metal part.

.NET Syntax

Visual Basic (Declaration)

```
Function GetFeatures() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim value As System.Object  
  
value = instance.GetFeatures()
```

C#

```
System.object GetFeatures()
```

C++/CLI

```
System.Object^ GetFeatures();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of [features](#) in this body in a sheet metal part

Visual Basic for Applications (VBA) Syntax

See [Body2::GetFeatures](#).

Example

[Get Features of Multibody Sheet Metal Part \(C#\)](#)
[Get Features of Multibody Sheet Metal Part \(VB.NET\)](#)
[Get Features of Multibody Sheet Metal Part \(VBA\)](#)

Remarks

The features of a body in a multibody sheet metal part are located in the solid bodies folder in the FeatureManager design tree.

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::GetFeatureCount Method](#)
[IBody2::IGetFeatures Method](#)
[IBodyFolder Interface](#)
[IBodyFolder::GetBodies Method](#)
[IBodyFolder::IGetBodies Method](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetFirstFace Method (IBody2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : GetFirstFace Method (IBody2)

Finds the first face in a body and returns the face.

.NET Syntax

Visual Basic (Declaration)

```
Function GetFirstFace() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim value As System.Object  
  
value = instance.GetFirstFace()
```

C#

```
System.object GetFirstFace()
```

C++/CLI

```
System.Object^ GetFirstFace();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

First face in the body

Visual Basic for Applications (VBA) Syntax

See [Body2::GetFirstFace](#).

Example

[Get Part Bounding Box \(VBA\)](#)
[Get Material Properties \(VBA\)](#)

[Traverse Bodies \(C++\)](#)
[Set Body for View \(C#\)](#)
[Set Body for View \(VB.NET\)](#)
[Set Body for View \(VBA\)](#)
[Delete Blended Faces \(C#\)](#)
[Delete Blended Faces \(VB.NET\)](#)
[Delete Blended Faces \(VBA\)](#)

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::GetFaceCount Method](#)
[IBody2::EnumFaces Method](#)
[IBody2::IGetFirstFace Method](#)
[IBody2::DeleteBlends2 Method](#)
[IBody2::GetFaces Method](#)
[IBody2::IGetFaces Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetFirstSelectedFace Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : GetFirstSelectedFace Method (IBody2)

Gets the first selected face on this body. For use with [IBody2::GetProcessedBodyWithSelfFace](#) and intended for IGES routines.

.NET Syntax

Visual Basic (Declaration)

```
Function GetFirstSelectedFace() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim value As System.Object  
  
value = instance.GetFirstSelectedFace()
```

C#

```
System.object GetFirstSelectedFace()
```

C++/CLI

```
System.Object^ GetFirstSelectedFace();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

First selected face

Visual Basic for Applications (VBA) Syntax

See [Body2::GetFirstSelectedFace](#).

Remarks

Do not use this method for general selection handling. If you want to handle items selected by the user or items selected with [IModelDocExtension::SelectByID2](#), use [ISelectionMgr::GetSelectedObject6](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::IGetFirstSelectedFace Method](#)
[IBody2::GetNextSelectedFace Method](#)
[IBody2::GetSelectedFaceCount Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetGraphicsBody Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : GetGraphicsBody Method (IBody2)

Gets the graphics body associated with this body.

.NET Syntax

Visual Basic (Declaration)

```
Function GetGraphicsBody() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IBody2
Dim value As System.Object

value = instance.GetGraphicsBody()
```

C#

```
System.object GetGraphicsBody()
```

C++/CLI

```
System.Object^ GetGraphicsBody();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[IGraphicsBody](#)

Visual Basic for Applications (VBA) Syntax

See [Body2::GetGraphicsBody](#).

Example

See the [IFacet](#) examples.

Remarks

This method is valid only if [IBody2::IsGraphicsBody](#) is true.

See Also

[IBody2 Interface](#)
[IBody2 Members](#)

Availability

SOLIDWORKS 2022 FCS, Revision Number 30

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetIgesErrorCode Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : GetIgesErrorCode Method (IBody2)

Gets the current IGES error code.

.NET Syntax

Visual Basic (Declaration)

```
Function GetIgesErrorCode( _  
    ByVal Index As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim Index As System.Integer  
Dim value As System.Integer  
  
value = instance.GetIgesErrorCode(Index)
```

C#

```
System.int GetIgesErrorCode(  
    System.int Index  
)
```

C++/CLI

```
System.int GetIgesErrorCode(  
& System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index
Position of the error within the current list of errors

Return Value

IGES error code

Visual Basic for Applications (VBA) Syntax

See [Body2::GetIgesErrorCode](#).

Remarks

This method is intended for use during IGES processing only.

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::GetIgesErrorCount Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetIgesErrorCount Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : GetIgesErrorCount Method (IBody2)

Gets the number of errors encountered while running an IGES routine.

.NET Syntax

Visual Basic (Declaration)

```
Function GetIgesErrorCount() As System.Integer
```

```
Dim instance As IBody2
Dim value As System.Integer

value = instance.GetIgesErrorCount()
```

C#

```
System.int GetIgesErrorCount()
```

C++/CLI

```
System.int GetIgesErrorCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of errors encountered while running an IGES routine

Visual Basic for Applications (VBA) Syntax

See [Body2::GetIgesErrorCount](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::GetIgesErrorCode Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetIntersectionEdges Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : GetIntersectionEdges Method (IBody2)

Obsolete. Superseded by [IBody2::GetIntersectionEdges2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetIntersectionEdges( _
    ByVal ToolBodyIn As System.Object _
) As System.Object
```

Visual Basic (Usage)

 Copy Code

```
Dim instance As IBody2
Dim ToolBodyIn As System.Object
Dim value As System.Object

value = instance.GetIntersectionEdges(ToolBodyIn)
```

C#

```
System.object GetIntersectionEdges(
    System.object ToolBodyIn
)
```

C++/CLI

```
System.Object^ GetIntersectionEdges(
    & System.Object^ ToolBodyIn
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ToolBodyIn
Temporary body used to perform the intersection

Return Value

Array of edges

Visual Basic for Applications (VBA) Syntax

See [Body2::GetIntersectionEdges](#).

Remarks

This method imprints a set of edges on both of these temporary bodies and returns the edges in an unordered list as shown below. The total number of edges in this array is twice the value returned from [IBody2::GetIntersectionEdgeCount](#).

[*Edge1imprintedOnTarget*, *Edge1imprintedOnTool*, *Edge2imprintedOnTarget*, *Edge2imprintedOnTool*]

where the target body is the IBody2 object used to call this method and the tool body is passed into this method as the first argument.

If the two bodies are in an assembly, then IBody2::GetIntersectionEdges generates a list of the intersection edges between the two bodies. To do this, the second body must be transformed in its coordinate space so that it is positioned the same with respect to the first body as it is in assembly space.

To align the two bodies before calling IBody2::GetIntersectionEdges or IBody2::GetIntersectionEdgeCount, calculate the transformation from the first body to the second body and set this as the transform for the second body using [IBody2::ApplyTransform](#).

In the case of a tangency condition (for example, a planar face contacting the cylindrical face of a cylinder), this method returns a single edge along the tangency.

You might also find that [IBody2::Operations2](#) provides an adequate solution. This method intersects a sheet body with a target body.

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::GetIntersectionEdges Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetIntersectionEdges2 Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : GetIntersectionEdges2 Method (IBody2)

Gets the edges resulting from the intersection of the specified tool body and this body.

.NET Syntax

Visual Basic (Declaration)

```
Function GetIntersectionEdges2( _  
    ByVal ToolBodyIn As System.Object, _  
    ByVal AddFaceIds As System.Boolean _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim ToolBodyIn As System.Object  
Dim AddFaceIds As System.Boolean  
Dim value As System.Object  
  
value = instance.GetIntersectionEdges2(ToolBodyIn, AddFaceIds)
```

C#

```
System.Object GetIntersectionEdges2(  
    System.Object ToolBodyIn,  
    System.bool AddFaceIds  
)
```

C++/CLI

```
System.Object^ GetIntersectionEdges2(  
& System.Object^ ToolBodyIn,  
& System.bool AddFaceIds  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ToolBodyIn
Temporary tool body used to perform the intersection with this body (see **Remarks**)
AddFaceIds
True to create IDs for any new faces, false to not (see **Remarks**)

Return Value

Array of edges

Visual Basic for Applications (VBA) Syntax

See [Body2::GetIntersectionEdges2](#).

Remarks

This method imprints a set of edges on both of the bodies and returns the edges in an unordered list as shown below. The total number of edges in the returned array is twice the value returned by [IBody2::IGetIntersectionEdgeCount](#).

[*Edge1imprintedOnTarget*, *Edge1imprintedOnTool*, *Edge2imprintedOnTarget*, *Edge2imprintedOnTool*]

where *Target* is this IBody2 object, and *Tool* is the IBody2 object passed into this method as ToolBodyIn.

If the two bodies are in an assembly, then this method generates a list of the intersection edges between the two bodies. Before calling this method, the two bodies must be aligned. To align the two bodies, calculate the transformation from the first body to the second body and then apply the transform to the second body using [IBody2::ApplyTransform](#).

In the case of a tangency condition (for example, a planar face contacting the cylindrical face of a cylinder), this method returns a single edge along the tangency.

You might also find that [IBody2::Operations2](#) provides an adequate solution when intersecting a sheet body with a target body.

AddFaceIds allows you to add IDs to any newly created faces. This may be useful if ToolBodyIn later becomes a permanent body, which can occur in macro features.

See Also

[IBody2 Interface](#)
[IBody2 Members](#)

Availability

SOLIDWORKS 2020 FCS, Revision Number 28.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetMassProperties Method (IBody2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : GetMassProperties Method (IBody2)

Gets the mass properties of this body.

.NET Syntax

Visual Basic (Declaration)

```
Function GetMassProperties( _
    ByVal Density As System.Double _
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IBody2
Dim Density As System.Double
Dim value As System.Object

value = instance.GetMassProperties(Density)
```

C#

```
System.object GetMassProperties(
    System.double Density
)
```

C++/CLI

```
System.Object^ GetMassProperties(
    & System.double Density
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Density
Density to use for the mass property calculations on this body

Return Value

Array of objects (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Body2::GetMassProperties](#).

Example

[Cut Body in Half Using Macro Feature \(VBA\)](#)
[Get Mass Properties of Assembly Component \(VBA\)](#)

Remarks

This method is intended for obtaining the mass properties of temporary objects but may also be used with the SOLIDWORKS [IBody2](#) object that is created by the user. To get the mass properties of the SOLIDWORKS IBody2 object created by the user, you can also use [IModelDocExtension::GetMassProperties](#), which uses the density currently set for the body's material.

The return value is an array of doubles as follows:

Solid body

```
[ CenterOfMassX, CenterOfMassY, CenterOfMassZ, Volume, Area, Mass(Volume*density), MomXX, MomYY, MomZZ, MomXY, MomZX, MomYZ ]
```

Sheet body

```
[ CenterOfMassX, CenterOfMassY, CenterOfMassZ, Area, Circumference, Mass(Area*density), MomXX, MomYY, MomZZ, MomXY, MomZX, MomYZ ]
```

Wire body

```
[ CenterOfMassX, CenterOfMassY, CenterOfMassZ, Length, 0, Mass(Length*density), MomXX, MomYY, MomZZ, MomXY, MomZX, MomYZ ]
```

You can use [ISldWorks::GetUserPreferenceDoubleValue](#) and swMaterialPropertyDensity to get the density used by the SOLIDWORKS part. Consistent with all other SOLIDWORKS API functions, units are metric unless otherwise specified.

SOLIDWORKS returns information (such as the center of mass) in relation to where the body was created. For example, if you create a block in a part file that is centered at (0,0,0), then SOLIDWORKS returns the center of mass as (0,0,0). If you then use this part at some random location within an assembly, and the body is obtained from the assembly component object using [IComponent2::GetBody](#), then SOLIDWORKS still returns the center of mass as (0,0,0). If you need to determine the body's center of mass in relation to the assembly coordinate system, you need to multiply the component transform and the center of mass coordinates (see [IComponent2::Transform2](#)).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::IGetMassProperties Method](#)

Availability

SOLIDWORKS 2001Plus FCS Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetMaterialIdName Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : GetMaterialIdName Method (IBody2)

Obsolete. Superseded by [IBody2::GetMaterialIdName2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetMaterialIdName() As System.String
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim value As System.String  
  
value = instance.GetMaterialIdName()
```

C#

```
System.string GetMaterialIdName()
```

C++/CLI

```
System.String^ GetMaterialIdName();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [Body2::GetMaterialIdName](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetMaterialIdName2 Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : GetMaterialIdName2 Method (IBody2)

Gets the material name for this body.

.NET Syntax

Visual Basic (Declaration)

```
Function GetMaterialIdName2() As System.String
```

Visual Basic (Usage)



```
Dim instance As IBody2
Dim value As System.String

value = instance.GetMaterialIdName2()
```

C#

```
System.string GetMaterialIdName2()
```

C++/CLI

```
System.String^ GetMaterialIdName2();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Material name

Visual Basic for Applications (VBA) Syntax

See [Body2::GetMaterialIdName2](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::SetMaterialIdName2 Method](#)
[IBody2::GetMaterialUserName2 Method](#)
[IBody2::SetMaterialUserName2 Method](#)
[IBody2::GetMaterialPropertyName Method](#)
[IBody2::GetMaterialPropertyValuesForFace Method](#)
[IBody2::RemoveMaterialProperty Method](#)
[IBody2::RemoveMaterialProperty Method](#)
[IBody2::SetMaterialProperty Method](#)
[IBody2::MaterialPropertyValues2 Property](#)
[IBody2::MaterialPropertyValues2 Property](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetMaterialPropertyName Method (IBody2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : GetMaterialPropertyName Method (IBody2)

Gets the names of the material database and the material for the specified configuration.

.NET Syntax

Visual Basic (Declaration)

```
Function GetMaterialPropertyName( _
    ByVal ConfigName As System.String, _
    ByRef Database As System.String _
) As System.String
```

Visual Basic (Usage)



```
Dim instance As IBody2
Dim ConfigName As System.String
Dim Database As System.String
Dim value As System.String

value = instance.GetMaterialPropertyName(ConfigName, Database)
```

C#

```
System.string GetMaterialPropertyName(
    System.string ConfigName,
```

```
    out System.string Database
)
```

C++/CLI

```
System.String^ GetMaterialPropertyName(
& System.String^ ConfigName,
& [Out] System.String^ Database
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ConfigName
Name of configuration (see **Remarks**)
Database
Name of material database

Return Value

Name of material

Visual Basic for Applications (VBA) Syntax

See [Body2::GetMaterialPropertyName](#).

Example

[Set Material \(C#\)](#)
[Set Material \(VB.NET\)](#)
[Set Material \(VBA\)](#)

Remarks

This method gets the name of the material database and an entry in the material database that you need for the XML lookup. Open the database as an XML stream and work with the XML API to get the data.

If...	Then...
Only the Default configuration exists for the part	Use "" for ConfigName
A material was not applied to the configuration	Database name and return value are blank

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::GetMaterialIdName2 Method](#)
[IBody2::GetMaterialUserName2 Method](#)
[IBody2::IGetMaterialPropertyValuesForFace Method](#)
[IBody2::IRemoveMaterialProperty Method](#)
[IBody2::RemoveMaterialProperty Method](#)
[IBody2::SetMaterialIdName2 Method](#)
[IBody2::SetMaterialUserName2 Method](#)
[IBody2::IMaterialPropertyValues2 Property](#)
[IBody2::MaterialPropertyValues2 Property](#)
[IBody2::SetMaterialProperty Method](#)
[IBody2::HasMaterialPropertyValues Method](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetMaterialUserName Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : GetMaterialUserName Method (IBody2)

Obsolete. Superseded by [IBody2::GetMaterialUserName2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetMaterialUserName() As System.String
```

Visual Basic (Usage)



```
Dim instance As IBody2
Dim value As System.String

value = instance.GetMaterialUserName()
```

C#

```
System.string GetMaterialUserName()
```

C++/CLI

```
System.String^ GetMaterialUserName();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [Body2::GetMaterialUserName](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetMaterialUserName2 Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : GetMaterialUserName2 Method (IBody2)

Gets the material name for this body; the material name is visible to the user.

.NET Syntax

Visual Basic (Declaration)

```
Function GetMaterialUserName2() As System.String
```

Visual Basic (Usage)



```
Dim instance As IBody2
Dim value As System.String

value = instance.GetMaterialUserName2()
```

C#

```
System.string GetMaterialUserName2()
```

C++/CLI

```
System.String^ GetMaterialUserName2();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Material name

Visual Basic for Applications (VBA) Syntax

See [Body2::GetMaterialUserName2](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::SetMaterialUserName2 Method](#)
[IBody2::SetMaterialIdName2 Method](#)
[IBody2::GetMaterialIdName2 Method](#)
[IBody2::GetMaterialPropertyName Method](#)
[IBody2::GetMaterialPropertyValuesForFace Method](#)
[IBody2::RemoveMaterialProperty Method](#)
[IBody2::SetMaterialProperty Method](#)
[IBody2::MaterialPropertyValues2 Property](#)
[IBody2::MaterialPropertyValues2 Property](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetMeshBody Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : GetMeshBody Method (IBody2)

Gets the mesh body associated with this body.

.NET Syntax

Visual Basic (Declaration)

```
Function GetMeshBody() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim value As System.Object  
  
value = instance.GetMeshBody()
```

C#

```
System.Object GetMeshBody()
```

C++/CLI

```
System.Object^ GetMeshBody();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[IMeshBody](#)

Visual Basic for Applications (VBA) Syntax

See [Body2::GetMeshBody](#).

Example

See the [IFacet](#) examples.

Remarks

This method is valid only if [IBody2::IsMeshBody](#) is true.

See Also

[IBody2 Interface](#)
[IBody2 Members](#)

Availability

SOLIDWORKS 2022 FCS, Revision Number 30

GetMiddleSurface Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : GetMiddleSurface Method (IBody2)

Inserts a midsurface in a body.

.NET Syntax

Visual Basic (Declaration)

```
Function GetMiddleSurface( _
    ByVal PlacementPercentage As System.Double, _
    ByRef Face1List As System.Object, _
    ByRef Face2List As System.Object, _
    ByRef Thickness As System.Object, _
    ByRef MiddleSurfaceBody As System.Object _
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IBody2
Dim PlacementPercentage As System.Double
Dim Face1List As System.Object
Dim Face2List As System.Object
Dim Thickness As System.Object
Dim MiddleSurfaceBody As System.Object
Dim value As System.Integer

value = instance.GetMiddleSurface(PlacementPercentage, Face1List, Face2List, Thickness,
MiddleSurfaceBody)
```

C#

```
System.int GetMiddleSurface(
    System.double PlacementPercentage,
    out System.object Face1List,
    out System.object Face2List,
    out System.object Thickness,
    out System.object MiddleSurfaceBody
)
```

C++/CLI

```
System.int GetMiddleSurface(
    & System.double PlacementPercentage,
    & [Out] System.Object^ Face1List,
    & [Out] System.Object^ Face2List,
    & [Out] System.Object^ Thickness,
    & [Out] System.Object^ MiddleSurfaceBody
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PlacementPercentage

Position where to insert the midsurface in the body; value of 50 indicates to insert the surface in the middle

Face1List

Array of faces into which the midsurface will be or was inserted; you can provide an array of faces for this parameter and Face2List; if you do not, then the faces are determined internally

Face2List

Array of faces into which the midsurface will be or was inserted; you can provide an array of faces for this parameter and Face1List; if you do not, then the faces are determined internally

Thickness

Array containing minimum and maximum thickness of sheet metal for pairs of faces

MiddleSurfaceBody

[Middle surface body](#)

Return Value

0 if no errors; -1 if errors

Visual Basic for Applications (VBA) Syntax

See [Body2::GetMiddleSurface](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::GetMiddleSurface Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetNextSelectedFace Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : GetNextSelectedFace Method (IBody2)

Gets the next selected face. For use with [IBody2::GetProcessedBodyWithSelfFace](#) and intended for IGES routines.

.NET Syntax

Visual Basic (Declaration)

```
Function GetNextSelectedFace() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim value As System.Object  
  
value = instance.GetNextSelectedFace()
```

C#

```
System.Object GetNextSelectedFace()
```

C++/CLI

```
System.Object^ GetNextSelectedFace();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Selected face

Visual Basic for Applications (VBA) Syntax

See [Body2::GetNextSelectedFace](#).

Remarks

Do not use this method for general selection handling. If you want to handle items selected by the user or items selected with [IModelDocExtension::SelectByID2](#), use [ISelectionMgr::GetSelectedObject6](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::GetNextSelectedFace Method](#)
[IBody2::GetFirstSelectedFace Method](#)
[IBody2::GetSelectedFaceCount Method](#)
[IBody2::GetProcessedBodyWithSelfFace Method](#)
[IBody2::GetSelectedFaces Method](#)
[IBody2::GetFirstSelectedFace Method](#)
[IBody2::GetProcessedBodyWithSelfFace Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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GetOriginalPatternedBody Method (IBody2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : GetOriginalPatternedBody Method (IBody2)

Gets the original body from this patterned body. Also gets the transformation of this body with respect to the original body.

.NET Syntax

Visual Basic (Declaration)

```
Function GetOriginalPatternedBody( _  
    ByRef Xform As MathTransform _  
) As Body2
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim Xform As MathTransform  
Dim value As Body2  
  
value = instance.GetOriginalPatternedBody(Xform)
```

C#

```
Body2 GetOriginalPatternedBody(  
    out MathTransform Xform  
)
```

C++/CLI

```
Body2^ GetOriginalPatternedBody(  
& [Out] MathTransform^ Xform  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Xform
[IMathTransform](#)

Return Value

[IBody2](#)

Visual Basic for Applications (VBA) Syntax

See [Body2::GetOriginalPatternedBody](#).

Example

[Get Original Body from Pattern Body \(VBA\)](#)
[Get Original Body from Pattern Body \(VB.NET\)](#)
[Get Original Body from Pattern Body \(C#\)](#)

See Also

[IBody2 Interface](#)
[IBody2 Members](#)

Availability

SOLIDWORKS 2011 SP01, Revision Number 19.1

SOLIDWORKS API Help

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GetProcessedBody Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : GetProcessedBody Method (IBody2)

Obsolete. Superseded by [IBody2::GetProcessedBody2](#).

Visual Basic (Declaration)

```
Function GetProcessedBody() As System.Object
```

Visual Basic (Usage)

```
Dim instance As IBody2
Dim value As System.Object

value = instance.GetProcessedBody()
```

C#

```
System.Object GetProcessedBody()
```

C++/CLI

```
System.Object^ GetProcessedBody();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [Body2::GetProcessedBody](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)

SOLIDWORKS API Help

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GetProcessedBody2 Method (IBody2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : GetProcessedBody2 Method (IBody2)

Pre-processes the geometry of the body using the specified parameters.

Visual Basic (Declaration)

```
Function GetProcessedBody2( _
    ByVal MaxUAngle As System.Double, _
    ByVal MaxVAngle As System.Double _
) As Body2
```

Visual Basic (Usage)

```
Dim instance As IBody2
Dim MaxUAngle As System.Double
Dim MaxVAngle As System.Double
Dim value As Body2

value = instance.GetProcessedBody2(MaxUAngle, MaxVAngle)
```

C#

```
Body2 GetProcessedBody2(
    System.double MaxUAngle,
    System.double MaxVAngle
)
```

C++/CLI

```
Body2^ GetProcessedBody2(
    & System.double MaxUAngle,
```

```
& System.double MaxVAngle
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

MaxUAngle
Maximum U angle

MaxVAngle
Maximum V angle

Return Value

Pointer to the [body](#); this body is a processed copy of the body for this part

Visual Basic for Applications (VBA) Syntax

See [Body2::GetProcessedBody2](#).

Example

[Process Body \(C#\)](#)
[Process Body \(VB.NET\)](#)
[Process Body \(VBA\)](#)

Remarks

Pre-processing (for example, IGES) is sometimes necessary for exporting to systems that have difficulty with periodic conditions.

Faces that lie on periodic surfaces that are not closed in the periodic direction might have bounds returned by [IFace2::GetUVBounds](#) that are on the periodic seam. Use [IModeler::SplitFaceOnParam](#) if the application needs to ensure that the face bounds lie completely on the surface bounds

This method does not split at a seam unless the face is closed in that parameter. Splitting only occurs if the face has a parametric extent greater than the specified pitch.

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::IGetProcessedBodyWithSelFace Method](#)
[IBody2::IGetProcessedBodyWithSelFace Method](#)
[IPartDoc::IGetProcessedBody2 Method](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

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GetProcessedBodyWithSelFace Method (IBody2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : GetProcessedBodyWithSelFace Method (IBody2)

Gets a processed body.

.NET Syntax

Visual Basic (Declaration)

```
Function GetProcessedBodyWithSelFace() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IBody2
Dim value As System.Object

value = instance.GetProcessedBodyWithSelFace()
```

C#

```
System.object GetProcessedBodyWithSelFace()
```

C++/CLI

```
System.Object^ GetProcessedBodyWithSelFace();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Body, which is a copy of the body for this part

Visual Basic for Applications (VBA) Syntax

See [Body2::GetProcessedBodyWithSelFace](#).

Example

[Get Selected Faces on Processed Body \(C#\)](#)
[Get Selected Faces on Processed Body \(VB.NET\)](#)
[Get Selected Faces on Processed Body \(VBA\)](#)

Remarks

This method requires that you preselect one or more faces.

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::GetProcessedBody Method](#)
[IBody2::IGetProcessedBodyWithSelFace Method](#)
[GetFirstSelectedFace Method](#)
[IBody2::GetNextSelectedFace Method](#)
[IBody2::GetSelectedFaces Method](#)
[IBody2::IGetFirstSelectedFace Method](#)
[IBody2::IGetNextSelectedFace Method](#)
[IBody2::IGetSelectedFaces Method](#)
[IBody2::GetSelectedFaceCount Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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GetPropertyExtension Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : GetPropertyExtension Method (IBody2)

Obsolete. Superseded by [IBody2::GetPropertyExtension2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetPropertyExtension( _  
    ByVal ID As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim ID As System.Integer  
Dim value As System.Object  
  
value = instance.GetPropertyExtension(ID)
```

C#

```
System.object GetPropertyExtension(  
    System.int ID  
)
```

C++/CLI

```
System.Object^ GetPropertyExtension(  
& System.int ID  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ID

Visual Basic for Applications (VBA) Syntax

See [Body2::GetPropertyExtension](#).

See Also

[IBody2 Interface](#)
[IBody2 Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetPropertyExtension2 Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : GetPropertyExtension2 Method (IBody2)

Gets the specified property extension on this body.

.NET Syntax

Visual Basic (Declaration)

```
Function GetPropertyExtension2( _  
    ByVal ID As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim ID As System.Integer  
Dim value As System.Object  
  
value = instance.GetPropertyExtension2(ID)
```

C#

```
System.object GetPropertyExtension2(  
    System.int ID  
)
```

C++/CLI

```
System.Object^ GetPropertyExtension2(  
& System.int ID  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ID

(Size of the array returned by [IBody2::AddPropertyExtension2](#)) - (Position of the property extension in the array)

Return Value

Value of the property extension

Visual Basic for Applications (VBA) Syntax

See [Body2::GetPropertyExtension2](#).

Remarks

IBody2::AddPropertyExtension2 adds property extensions to a last-in-first-out, 1-based array and returns the size of that array.

NOTE: SOLIDWORKS recommends that you use the [IAttribute](#), [IAttributeDef](#), and [IParameter](#) interfaces instead of this method. These three interfaces provide more flexibility.

See Also

[IBody2 Interface](#)
[IBody2 Members](#)
[IBody2::ResetPropertyExtension2 Method](#)

SOLIDWORKS API Help

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GetSafeBody Method (IBody2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IBody2 Interface](#) : GetSafeBody Method (IBody2)

Not implemented.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSafeBody() As Body2
```

Visual Basic (Usage)



```
Dim instance As IBody2  
Dim value As Body2  
  
value = instance.GetSafeBody()
```

C#

```
Body2 GetSafeBody()
```

C++/CLI

```
Body2^ GetSafeBody();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Pointer to the [IBody2](#) object

NOTE: This method is not yet implemented; thus, it will currently always return NULL.

Visual Basic for Applications (VBA) Syntax

See [Body2::GetSafeBody](#).

Remarks

To determine if a body is safe, use the [IBody2::IsSafe](#) property. This property is read-only and does not persist from session to session.

See Also

[IBody2 Interface](#)
[IBody2 Members](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0